JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
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An Autonomous Institute, with NAAC "A" Grade
Department of CSE
Education to Eternity
"A Place to Learn ; A Chance to Grow"
Session 2024-25

MSE EXAMINATION

Course: B. Tech. in Computer Science & Engineering **Subject Code**:CS3T002

Subject: Operating System **Year/Semester:** 3rd SEM CSE

Max Marks: 20 **Date:** 5 /9 /2024 **Duration:- 1 Hr.**

Instructions to the Students:
1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students will be able to:
1 Understand the concepts, core structure of Operating Systems.
2 Learn how processes are managed and scheduled by the OS, including understanding process states, context switching, and various CPU scheduling algorithms.
3 Develop an understanding of synchronization mechanisms like semaphores, monitors, and the handling of deadlocks through detection, prevention, and avoidance strategies.
4. Acquire knowledge of memory management concepts such as paging, segmentation, virtual memory, and how the OS allocates, manages, and deallocates memory.
5 Analyse various device and resource management techniques for timesharing

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define a Thread	L1/CO2	1
Q.2	List the common synchronization mechanisms used in operating systems.	L1/CO2	1
Q.3	State the term starvation and deadlock.	L1/CO3	1
Q.4	Quote context switching in process synchronization.	L1/CO3	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Illustrate the advantages and disadvantages of FCFS algorithm.	L2/CO2	2
Q.6	Explain Classical Synchronization problems in short.	L2/CO3	2
Q.7	Compare semaphores and monitors for process synchronization.	L4/CO3	2
Q.8	Describe how the Round Robin scheduling algorithm works.	L2/CO2	2

SECTION C

1. **Solve any two questions.**

Q.9	Evaluate a given scheduling algorithm based on its ability to meet various scheduling criteria in a multi-user environment."	L5/CO2	4
Q.10	Consider the set of 5 processes whose arrival time and burst time are given below-	L3/CO2	4

Process Id	Arrival time	Burst time	Priority
P1	0	4	2
P2	1	3	3
P3	2	1	4
P4	3	5	5
P5	4	2	5

If the CPU scheduling policy is priority non-pre-emptive, calculate the average waiting time and average turnaround time.

Q.11 Analyse the conditions under which a deadlock might occur, and determine how these conditions can be mitigated." L4/CO3 4

********** **********All the best********** **********